The Silicon-Power Gap

A variant equity thesis on the 2028 AI compute buildout

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The Setup

The 2028 AI compute buildout is a hyperscaler capex cycle approaching $700 billion for 2026, and investor exposure is concentrated on one premise: that NVIDIA chip supply is the binding constraint. This note argues the consensus underweights the power-delivery bottleneck. The binding input sits upstream of the chip, in a different industry, where the manufacturers' order books are public. If that is right, the AI timetable is set by three transformer manufacturers, not by NVIDIA.

The Variant: It Is the Transformer, Not the Chip

The binding constraint on 2028 AI compute is the large oil-cooled grid transformer, the apparatus at the head of the power-delivery chain that steps high-voltage utility power down to facility-side medium-voltage service, ahead of the UPS, PDU, and server power-supply stages. The case rests on four facts.

Element	What the data shows
Unit economics	Large transformer class, 100 to 200+ MVA; the modeled unit is 100 MVA at about $9.2M and roughly 150 tons; roughly 128 to 144 weeks on average, with GSU and substation units quoted as far out as 210 weeks of quoted order-to-delivery lead time.
Supply structure	A three-supplier oligopoly in OECD HV, GSU-class supply at data-center delivery windows: Hitachi Energy, Siemens Energy, GE Vernova.
Interconnection queue	The national median from interconnection request to commercial operation runs beyond five years for projects energized in 2025, against 4.5 years for the 2024 cohort (Rand et al. 2025, LBNL Queued Up). A generation/storage proxy: large-load data-center interconnection varies by utility but faces the same power-delivery and equipment bottlenecks.
Stacked inputs	Grain-oriented electrical steel from a handful of mills, plus a multi-year apprenticeship pipeline for the certified high-voltage winders the suppliers cannot expand without.

The implication is direct: an operator can buy every NVIDIA chip it wants. If it did not place its transformer orders by 2024, it cannot turn those chips on until 2029 or 2030. Operators that ordered in 2024 hold inventory competitors cannot replicate within the same window without a large premium or execution risk. Hitachi Energy's order backlog corroborates the supplier-side shortage: about $31 billion at the end of FY2023 to $57.9 billion at the end of FY2025, an eighty-seven percent expansion in two years. That is the shortage showing up in a public-company revenue runway, not in a forecast.

The Geographic Asymmetry: Miami vs. Helsinki

Cellulose paper insulation aging accelerates with winding hot-spot temperature under an Arrhenius relation, codified for transformers in IEEE Standard C57.91, which sets the aging-acceleration factor against a reference hot-spot:

$$F_{AA} = exp[(E_A/R)(1/\theta_{ref} - 1/\theta_H)]$$

where EA/R is about 15,000 K for mineral-oil-impregnated paper insulation, θref is the 383 K (110 °C) reference hot-spot, and θH is the site winding hot-spot. The same physics that tightens the constraint in the Sun Belt loosens it in the Nordic region.

Metric	Miami (annual mean ambient 25.3 °C)	Helsinki (annual mean ambient 6.1 °C)
Relative aging index (Northern Virginia = 1.00)	2.72	0.45
Expected life, competing risks	18.6 yr	36.5 yr
Free-cooling availability (no chiller)	6% of year	89% of year
Reference data-center PUE	1.42 (illustrative Miami air-cooled case)	1.10 (Google Hamina)

The hyperscalers have already revealed where this points. Google has operated a major facility at Hamina, Finland since 2011 (site acquired 2009). Meta operates Node Pole at Luleå inside the Arctic Circle. Nebius, formerly Yandex's international business, sells waste heat into Finnish district heating at Mäntsälä. These are infrastructure-economics decisions, and the advantage is not consistently capitalized in public mid-tier operator models.

The Valuation: Each Slot Is a Real Option

Each secured 2024-vintage transformer slot behaves like a real option on accelerated AI compute deployment. I value it inside the Trigeorgis (1996) real-options framework, with a Black-Scholes-Merton parameterization to anchor the order of magnitude:

$$C = S_0 \cdot N(d_1) - K \cdot e^{-rT} \cdot N(d_2)$$

$$d_1 = [\ln(S_0/K) + (r + \sigma^2/2)T] / (\sigma \sqrt{T}), \qquad d_2 = d_1 - \sigma \sqrt{T}$$

S0 is the risk-adjusted present value of the scarcity rent the slot pulls forward for the landlord: about $0.40M per MW per year of scarcity rent, times 75 MW, times roughly two years, arriving at $30M at t = 1.5 and t = 2.5 and discounted at the risk-free rate to $55.1M. K is the carrying cost of pulling the fit-out forward two years, 450 x (1 - e^(-0.044 x 2)) = $37.9M, not the fit-out itself; earlier drafts paired a $61M acceleration increment against a $42M full build cost, which measured S and K on different objects. T = 1 year is the competitive-exclusion window to the delivery decision, and S0 embeds the multi-year rent window after exercise. The 35% volatility is borrowed from realized H100 spot rental volatility as a deliberately high proxy. A practitioner reviewer argued the GPU-spot object is the wrong one; that criticism is accepted, and it runs in the thesis's favour, because contracted landlord scarcity rent is less volatile and a lower volatility pushes the option deeper in-the-money; the option is deep in the money, so intrinsic value, not volatility, drives the result, and across a 15% to 60% volatility band the per-slot value moves only from $18.8M to $22.6M and the order-book value holds between about $301M and $361M.

- Per slot (75 MW): a $15M to $25M range, central illustrative value $19.7M.
- A 1.2 GW order book (16 slots): $315M gross; less the $8M liquidity discount and the $31M Sun Belt thermal liability, about $276M net. The +$24M correlated-constraint moat premium carried in earlier drafts is retired.

The thesis does not rest on the central numbers. It rests on three facts the order-of-magnitude calculation establishes: the option value is positive and material, it is material at the asset and order-book level, and it is asymmetric across geographies.

The Trade: A Two-Name Relative-Value Book (Both Legs Long)

Leg	Position	Hidden value	Driver
Long	Operator X (US-listed, transformer-rich)	about $276M	2024-vintage slot inventory and option value, net of the Sun Belt thermal liability. The compound regulatory, metallurgical and labor moat is described but is not separately capitalised.

Leg	Position	Hidden value	Driver
Long	Operator H (Nordic: Helsinki, Luleå, Stockholm)	about $104M	Thermal and free-cooling advantage, low-carbon-grid premium, a small waste-heat revenue line, and a qualitative NATO and EU jurisdictional premium for sovereign-AI and defense procurement.

The Carbon Asymmetry and the Disclosure Catalyst

The long legs split on a second axis the sell-side ignores: carbon claims. Sun Belt operators publish carbon-neutral and 24/7 carbon-free-energy claims their physics does not support (PUE about 1.42, free cooling 6% of the year, an ERCOT grid near 375 gCO2/kWh); Nordic operators make the same claims on grids at roughly 32 gCO2/kWh operational (Finland, 2024) and 35 to 80 on lifecycle estimates with free cooling 89% of the year. Same claim, different substance. This is the audit-threat framing of Lyon and Maxwell (2011): firms disclose their true positives and withhold their negatives, and the withheld negative surfaces only if an audit arrives. The framing supplies a taxonomy of enforcement channels, not a timetable; the auditing party here would be a strategic rival, whose choice among channels makes the audit probability endogenous, and possibly zero.

The reporting mechanism is live; the exposure mechanism is not yet. The EU Energy Efficiency Directive data-centre scheme (Commission Delegated Regulation 2024/1364) has collected PUE, renewable-energy factor, and waste-heat reuse from every EU facility above 500 kilowatts since September 2024, but facility-level KPIs are confidential by Article 5(5): the first aggregated release (April 2026) published only aggregates, 776 facilities and 6.4 GW at a weighted average PUE of 1.36, on partial member-state coverage. The rival operator’s own KPIs are audited under the scheme while the laggard’s stay confidential, so the comparison the market needs is the piece nothing compels into the open until the pending second-phase rating scheme is adopted. CBAM certificate sales begin February 2027 for 2026 imports (Regulation (EU) 2025/2083), and CSRD wave-two reports arrive for financial year 2027 under the narrowed scope of Directive (EU) 2026/470, raising the cost of an unverified claim into the same window.

The conviction read splits: Medium on the mechanism, Low on timing inside the Wealth Management (1 to 3 year) window, until either a rival commits to an enforcement channel or the second-phase rating scheme is adopted and converts a confidential database into published comparability. It is the mechanism and the channel, not the magnitude, that the Nordic long leg is exposed to.

Operator X is a stylized public-equity profile; Operator H is a Nordic-exposure profile with no clean public pure play. Both legs are long: the Sun Belt thermal drag enters as a liability embedded inside Operator X, not a separate equity short. Combined hidden value is about $380 million the market has not yet capitalized; that lands within 5% of the approximately $400 million probability-weighted figure from a separate bear-case pass over the same inputs, a consistency check rather than an independent one. The legs are correlated under the disclosure catalyst: exposing Sun Belt carbon claims would simultaneously force recognition of the thermal liability the Operator X bridge already carries at full value.

Why This Matters

The same supply chain that gates AI compute also gates electrification: AI load competes with EV charging, heat pumps, and renewable interconnection for the same transformers, steel, and switchgear. The mispricing has a forced resolution: either the market reprices on the producer data, or the constraint binds at the deployment timeline and the operators without inventory disclose it.

The Next Migration: Off-Earth Compute

The next migration, off-Earth compute, is already a spending category (Starcloud, Google Suncatcher, the SpaceX AI1 satellite, China’s Three-Body constellation). It does not threaten the thesis; it confirms it.

Every escape the market imagines, behind-the-meter gas, SMRs, demand-side flexibility, orbit, relocates the bottleneck rather than clearing it, and each still terminates at a step-up transformer.

What Would Settle This

- **Confirms: sell-side initiates on Operator X or H citing secured transformer inventory, a transformer-naked Sun Belt operator guides down on power-constrained capex, or a hyperscaler discloses a transformer-bound delay. Falsifies: suppliers materially expand 200+ MVA capacity ahead of guidance, solid-state transformers reach hyperscale cost parity before 2028 (a low-probability tail), or interconnection queues clear to a sub-30-month median. Base case (no near-term resolution): the mispricing persists into 2028 and the trade compounds through deployment delays, the modal case in my scenario tree.**

Key Assumptions and Ranges

The per-slot valuation is a Trigeorgis-style heuristic; the $276M and $104M figures are modeled estimates inside ranges of $240M to $400M and $95M to $125M. Backlog data is from public earnings reports. The underlying paper has been read and critiqued in writing by senior practitioners, among them Alejo Czerwonko (UBS; Columbia SIPA), Gonzalo Binello (Schroders), Riccardo Bertocco (former BCG Power and Utilities), and Mariano Scandizzo (Insigneo); those critiques are addressed in the full paper, posted with its valuation model and correction log. Prepared with AI assistance: Claude models from Anthropic drafted the prose, which I directed and edited; the paper's authorship note sets out the division of labor.

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